CLSHA-YOLOV8: YOLOV8 BASED ON CROSS-LAYER SINGLE-HEAD ATTENTION FOR OBJECT DETECTION IN UAV IMAGES

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***Abstract*—A YOLOv8 object detection network based on cross-layer single-head attention (CLSHA-YOLOv8) is proposed to address the challenges of detecting small and occluded objects in UAV images. The network incorporates a cross-layer single-head attention (CLSHA) module, which efficiently allocates attention by guiding lower-layer feature maps through higher-layer feature maps and calculating single-head attention using partial channels. This design enhances the model's global context-awareness, facilitates the inference of occluded object locations, and effectively controls computational costs. Furthermore, the module improves the detailed representation of objects by introducing residual connections for specific channels in the low-level feature maps. This enhancement contributes to higher mAP for small objects. Experiments on the VisDrone2019 dataset demonstrate that the proposed network achieves enhanced performance in detecting small and occluded objects.**

***Index Terms*—object detection, UAV images, small objec, occlusion object, cross attention.**

I. Introduction

The integration of UAV with deep learning-based object detection techniques has emerged as a prominent research focus. By incorporating advanced object detection algorithms, UAV can autonomously analyze their surroundings, offering robust capabilities for real-time monitoring and decision-making. However, while expanding the potential applications of UAV, they also introduce unique challenges. Detecting small objects is a significant challenge in UAV imagery. High-altitude captures often result in objects image that are extremely small—sometimes just a few hundred or even tens of pixels. Such limited size causes a loss of critical details during feature extraction, making it difficult to acquire effective features and ultimately reducing detection accuracy. Furthermore, small objects are more susceptible to being overwhelmed by image noise or blending with their surroundings, increasing the likelihood of missed or false detections. Another challenge arises from the frequent occlusion observed in UAV images, where objects may be partially obscured by other objects or the background. This often renders parts of the objects invisible and causes boundary blurring. These factors further complicate the detection task and hinder the accurate identification of objects.

To address the challenges in small object detection and occlusion, scholars have proposed various improvements in different directions. For small object detection, multi-scale feature fusion methods are commonly employed to enhance performance. These methods integrate the rich detail information in low-level feature maps with the semantic richness of high-level feature maps, thereby improving both object localization and classification accuracy. For instance, Zhao et al. [1] designed a cross-layer asymmetric transformer to fuse information across different layers. Xia et al. [2] proposed a CNN-Transformer bidirectional fusion interaction module capable of multi-scale fusion between hierarchical features, enhancing the interaction of local information. Li et al. [3] developed a location refined feature pyramid network to enhance shallow location information extraction and promote fine-grained contextual interaction, enabling more effective localization of small objects in remote sensing images. Wang et al. [4] presented a tiny path fusion module that utilizes multi-scale feature map fusion and deformable convolution to improve the fusion of details in low-level feature maps.

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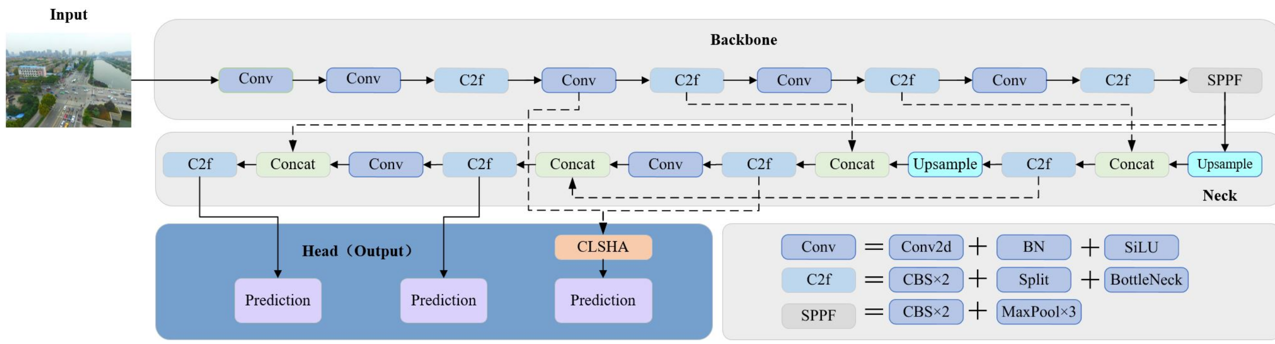


Fig. 1. The overall framework of our CLSHA-YOLOv8.

These multi-scale fusion methods have demonstrated their potential to improve the accuracy of small object detection, proving effective in addressing this challenge. For occluded object detection, many studies focus on modeling global context and leveraging contextual information to distinguish similar objects and compensate for missing object details. For example, TPH-YOLOv5 [5] introduced the transformer prediction head, MS-YOLOv7 [6] employed the swin transformer prediction head, and SCDNet [7] designed a scene context guided fusion module. These approaches use multi-head self-attention to model long-range spatial dependencies, effectively capturing global context information and enhancing the network's ability to identify overlapping and occluded objects. However, although the multi-head self-attention mechanism is highly effective at capturing global information, it comes with substantial computational overhead, significantly increasing model complexity and adversely affecting detection efficiency.

Inspired by previous studies, this paper proposes a YOLOv8 object detection network based on cross-layer single-head attention (CLSHA-YOLOv8). The CLSHA module is useded to preserve the advantages of multi-head attention in capturing long-range and complex contextual information while minimizing computational redundancy. Unlike traditional approaches, this module leverages a single-head attention mechanism, inspired by the Single-Head Vision Transformer (SHViT) [8], to address computational inefficiencies by applying attention computations to a subset of feature map channels. Despite utilizing only partial channel attention, the single-head approach enhances the model's global context-awareness by modelling long-range dependencies, which is particularly beneficial for inferring the locations of occluded objects. The CLSHA module differs from the structure of SHViT by utilizing both high-level and low-level feature maps as inputs, rather than relying on a single feature map. This design allows high-level semantic features to guide the allocation of attention across low-level detailed features, thereby enhancing the model's capability to preserve critical details essential for small-object detection. Additionally, the module introduces residual connectivity for selected channels, further reinforcing the retention of fine-grained details. This approach improves the model's capability to accurately detect small objects and handle occluded objects more effectively.

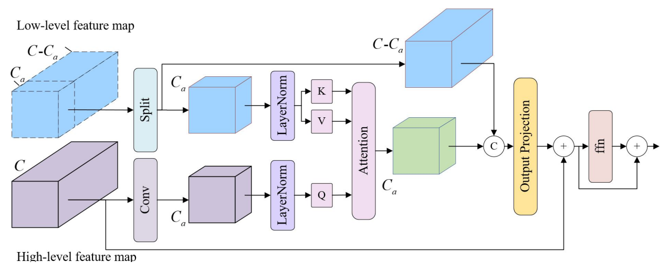


Fig. 2. CLSHA module structure.

The main contributions of this paper as follows :

(1)A novel CLSHA-YOLOv8 network is proposed to address the challenges of detecting small and occluded objects in UAV images.

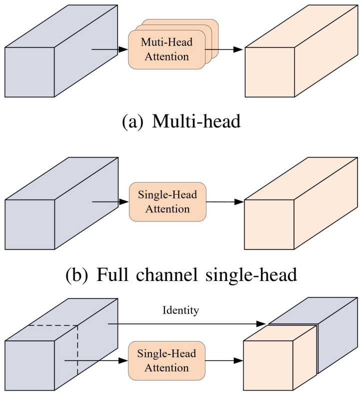
(2)To enhance detail preservation for small objects and model long-range dependencies, a cross-layer single-head attention (CLSHA) module is designed. This module improves detection accuracy for small and occluded objects while reducing computational redundancy.

(3)A novel CLSHA-YOLOv8 network is proposed for tackling the challenges of detecting small and occluded objects in UAV images.

II. Methodologye

YOLOv8 [9], a convolutional neural network-based object detection algorithm, demonstrates strong performance in many standard scenarios. However, it faces challenges in detecting tiny objects and effectively capturing global contextual information. To address this challenge and enhance the model's ability to extract global contextual information, this paper proposes CLSHA-YOLOv8, a network designed to improve the detection rate of small and occluded objects in UAV images. The structure of this network is illustrated in Fig.1.

Drawing inspiration from the SHViT [8] and incorporating the concept of cross-layer information fusion, this study proposes an innovative CLSHA module. Inspired by the single-head Vision Transformer (SHViT) [8] and leveraging the concept of cross-layer information fusion, this study proposes an innovative CLSHA module. The structure of this module is shown in Fig.2.



(a) Multi-head

(b) Full channel single-head

(c) Partial channel single-head

Fig. 3. Comparison of multi-attention and single-attention.

In this module, both high-level and low-level feature maps are used as inputs to the attention mechanism. High-level feature maps, enriched with semantic information, guide the attention allocation of low-level feature maps containing detailed information. This design significantly enhances the model's ability to perceive relationships among objects in densely occluded scenes, thereby improving detection performance. Furthermore, the CLSHA module reduces the computational complexity of the attention mechanism by restricting attention computations to a subset of channels. This approach also balances computational efficiency and the model's ability to extract global contextual information, avoiding substantial increases in computational overhead. The structure and implementation of the CLSHA module are detailed in Fig.2, which provides a comprehensive overview of its design.

To better illustrate the computational efficiency of the partial-channel single-head attention employed in the CLSHA module, some tests have made between multi-head attention, full-channel single-head attention, and the partial-channel single-head attention used in this studies. The comparison graph is shown in Fig.3. As depicted in the figure, multi-head attention requires multiple attention computations across the entire feature map, while full-channel single-head attention performs a single attention computation for the full channel. So in contrast, the partial-channel single-head attention used in this study computes attention for only a subset of the channels in the feature map, with the remaining channels directly concatenated with the computed feature map. Compared to both multi-head attention and full-channel single-head attention, the partial-channel single-head attention reduces computational complexity and memory consumption, ensuring that the model operates more efficiently while effectively capturing long-range dependencies.

III. Experiments

A. Dataset and Experimental Settings

For the experimental setup, model training is conducted using two GeForce RTX 2080 Ti GPUs with 11GB of VRAM each. Both the baseline model and our proposed model are implemented using the PyTorch 1.12.1 framework and trained with the stochastic gradient descent (SGD) optimization algorithm. The training parameters are configured as follows: the initial learning rate is set to 0.01, the final learning rate to 0.001, the momentum factor to 0.937, and the batch size is set to 2. The input image size for the VisDrone2019 dataset [10] is 960×960. The training process span 200 epochs. Additionally, no additional data augmentation techniques are applied to the models in this study beyond those used in the baseline model.

B. Results

The comparison results for the VisDrone2019 validation set are presented in TABLE I. The comparison models in the table include both classical object detection algorithms and more recent advanced algorithms from the past two years. Since the VisDrone2019 dataset is publicly available and has been officially delineated, the experimental data from different papers are highly comparable. Specifically, the experimental data for DTSSNet and ATSS are sourced from the paper [11] and [12], while the data for ARFENet are from the paper [4]. The results for both the baseline model and our model are derived from native experiments and ensure consistency in the hyperparameter settings.

From the experimental results in TABLE I, it is evident that our model surpasses the compared models in terms of mAP and mAP_{50} metrics. Although it falls slightly behind DTSSNet in the mAP_S metric, DTSSNet exhibits significantly lower mAP for medium and large object detection and incurs 1.7 times the computational cost of our model. These results clearly demonstrate the effectiveness of our proposed network in enhancing the mAP of UAV image object detection.

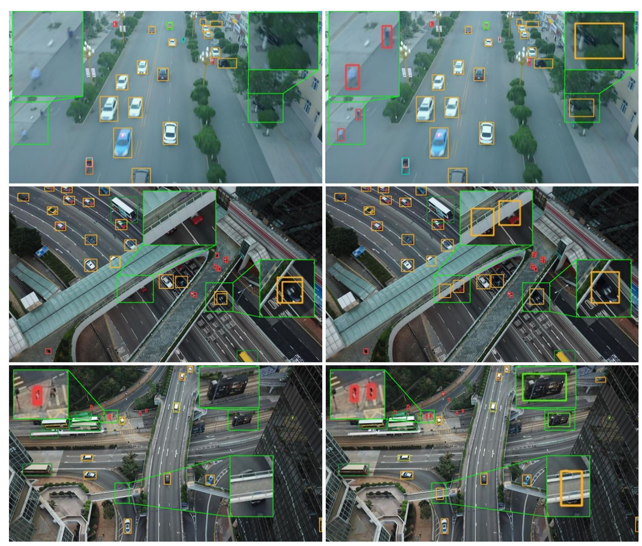

(a) Baseline (b) Ours

Fig. 4. Comparison of the baseline model and the our model on the VisDrone2019 test set.

To illustrate the differences between the baseline and our models in detecting small and occluded objects on the VisDrone2019 dataset, we compare their detection results, as

TABLE I
PERFORMANCE EVALUATION ON VISDRONE2019 VALIDATION SET.

model	mAP	mAP_{50}	mAP_S	mAP_M	mAP_L	$Params(M)$	$GFLOPs$
ATSS [12]	18.6%	31.7%	12.9%	25.9%	29.5%	10.3%	56.9%
YOLOv8-s [9]	27.5%	46.0%	16.3%	37.2%	42.0%	11.2%	**28.5%**
DTSSNet [11]	24.2%	39.9%	**17.5%**	32.3%	36.1%	**10.1%**	50.3%
ARFENet [4]	22.9%	39.0%	13.2%	31.4%	**45.3%**	10.4%	28.7%
Ours	**28.5%**	**47.3%**	17.1%	**38.3%**	44.2%	11.3%	29.6%

shown in Fig.4. In the first scene, the baseline model fails to detect a blurry, tiny pedestrian object and a vehicle occluded by foliage, whereas the our model successfully detects both. In the second scene, the baseline model misses two cars obscured by a bridge and produces duplicate detection for another occluded car, while our model accurately detects the two obscured cars without any duplication. In the third scene, the baseline model fails to detect a tiny pedestrian, a bus, and a car occluded by a bridge, whereas the proposed model successfully identifies all these objects. These results demonstrate that our method effectively enhances the detection accuracy of small and occluded objects.

TABLE II
RECALL AND mAP ABLATION EXPERIMENTAL RESULTS OF THE CLSHA MODULE AND ITS VARIANTS ON THE VISDRONE2019 VALIDATION SET.

model	R	mAP	mAP_{50}	mAP_S	mAP_M	mAP_L
+A	44.7%	27.7%	46.0%	16.7%	37.3%	40.9%
+B	43.5%	27.0%	45.1%	15.9%	36.6%	41.9%
+CLSHA	**45.2%**	**28.5%**	**47.3%**	**17.1%**	**38.3%**	**44.2%**

TABLE III
EXPERIMENTS OF THE CLSHA MODULE AND ITS VARIANTS ON THE VISDRONE2019 VALIDATION SET.

model	$Params(M)$	$GFLOPs$	FPS
+A	11.3	29.8	69
+B	11.3	29.6	76
+CLSHA	**11.3**	**29.6**	**76**

To analyze the role of partial channel attention and cross-layer input in the CLSHA module, we conducted experiments on the VisDrone2019 validation set, with the results presented in TABLE II and TABLE III. In this context, "+A" refers to replacing the single-head partial-channel attention in the CLSHA module with a single-head full-channel attention, which was then added to the baseline model, and "+B" indicates that the cross-layer feature map input was replaced with a single-layer feature map input, which was subsequently incorporated into the baseline model.

As shown in TABLE II, the CLSHA module outperforms both variants in recall and all mAP metrics. Compared to the single-head full-channel attention module, the CLSHA module improves recall by 0.5%, mAP and mAP_{50} by 0.8% and 1.3%, and mAP_S, mAP_M, and mAP_L by 0.4%, 1.0%, and 3.3%, respectively. This improvement stems from the CLSHA module's use of partial channels, which are directly spliced with the attention output, enhancing the retention of target details and improving detection accuracy in UAV images. When compared to the single-layer feature map input module, the CLSHA module boosts recall by 1.7%, mAP and mAP_{50} by 1.5% and 2.2%, and mAP_S, mAP_M, and mAP_L by 1.2%, 1.7%, and 2.3%, respectively. This advantage results from the CLSHA module's cross-layer inputs, which leverage high-level semantic information to guide low-level details, strengthening inter-layer interaction and improving detection accuracy.

As shown in TABLE III, in terms of detection efficiency, the CLSHA module is on par with the single-layer feature map input variant but outperforms the full-channel attention variant. Compared to the full-channel attention variant, the CLSHA module reduces overall computation by 0.2 GFLOPs, demonstrating that partial-channel single-head attention computation effectively minimizes computational redundancy and enhances operational efficiency.

IV. CONCLUSION

In this paper, we propose CLSHA-YOLOv8, which designs the CLSHA module to improve the detection of small and occluded objects in UAV images. The module employs cross-layer attention to establish long-range dependencies and enhance global contextual information, thereby improving the accuracy of occluded object detection. Additionally, it uses part-channel attention computation and direct splicing of unprocessed channels to retain detail information while reducing computational redundancy, enhancing the detection of small objects. Experimental results show that our method achieves the mAP of 28.5% and the mAP_{50} of 47.3% on the VisDrone2019 validation set, with only 0.1M increase in parameters and 1.4 GFLOPs increase in computation.

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